

Few-Shot Distillation for Real-time Surgical Phase Recognition under Label Scarcity

Minqing Zhang, Second B. Author, and Wu Yuan, *Senior Member, IEEE*

Abstract—As surgical phase recognition (SPR) enters the foundation model (FM) era, the FM paradigm itself exposes two clinical deficiencies: it relies on massive training data that is prohibitive to curate in surgery, and its heavy architectures cannot sustain the real-time inference that clinical deployment demands. Our pilot study further shows that, under few-shot SPR, generic FMs match or outperform specialized surgical FMs, indicating that the data deficiency can be sidestepped rather than satisfied. We therefore build on a generic visual prior and present a redundancy-aware framework that turns the dual redundancy of surgical videos—dominant background regions and repetitive steady-state frames—from an obstacle into an inductive bias under scarce supervision. First, to break the FM paradigm’s reliance on massive data, we introduce SPR-Teacher, a spatiotemporal adaptation tuning framework that adapts a frozen generic image FM to surgical video with only a handful of labeled videos: salient spatial token selection (SSTS) compresses spatial redundancy through a soft top-K saliency gate, and keyframe-prior temporal fusion (KPTF) compresses temporal redundancy by aggregating the most discriminative moments. Second, to cure the real-time deficiency, we propose an architecture-agnostic distillation strategy that compresses SPR-Teacher into SPR-Lite, a lightweight student family for arbitrary backbones: a cross-attention projection bridges heterogeneous student feature maps into the teacher’s token space, and a spatiotemporally-guided distillation transfers the teacher’s where- and when-to-look priors as dense supervision to compensate for the scarce labels. Comprehensive experiments on five SPR datasets, including four public benchmarks and one private clinical cohort, demonstrate that SPR-Teacher substantially outperforms state-of-the-art methods in few-shot scenarios, and SPR-Lite approaches

this performance with under 9 million parameters at over 25 FPS on a single CPU.

Index Terms—Surgical phase recognition, few-shot learning, knowledge distillation, real-time deployment.

I. INTRODUCTION

SURGICAL phase recognition (SPR), which identifies frame-wise surgical intentions from procedural videos, is foundational for intelligent operating rooms by supporting intraoperative decision support [1], real-time anomaly detection [2], and postoperative skill assessment [3]. Recently, the domain has shifted toward pre-training surgical foundation models (FMs) on expansive corpora [4]–[6], which demonstrate promising performance. The FM paradigm itself, however, carries two deficiencies that loom large in clinical practice. The first is its reliance on massive training data: curating expansive surgical corpora incurs prohibitive annotation costs, privacy risks, and procedural biases. The second is its inability to run in real time: the parameter explosion that accompanies scaling renders heavy architectures computationally infeasible for resource-constrained clinical hardware, while practical SPR demands real-time inference, as instant intraoperative feedback directly impacts patient safety. Consequently, overcoming the data hunger and the deployment latency of the FM paradigm remains a critical yet overlooked challenge for practical SPR.

The data deficiency, we argue, can be sidestepped rather than satisfied: generic visual FMs already provide rich pre-trained representations, and our systematic evaluation across five surgical datasets reveals that they match or outperform specialized surgical FMs in few-shot scenarios, despite lacking any surgical pretraining exposure (Sec. IV-D.1). We therefore build on a generic visual prior and propose SPR-Teacher, a redundancy-aware spatiotemporal adaptation framework that slashes the annotation requirement of SPR from massive corpora to a handful of labeled videos. Surgical videos exhibit a dual redundancy: background regions dominate the field, and steady-state frames repeat within each phase, diluting the already scarce supervision. Salient spatial token selection (SSTS) and keyframe-prior temporal fusion (KPTF) turn this redundancy into an inductive bias, filtering irrelevant background while preserving discriminative moments for effective video understanding. Consequently, SPR-Teacher breaks the FM paradigm’s reliance on massive training data.

This paragraph of the first footnote will contain the date on which you submitted your paper for review. It will also contain support information, including sponsor and financial support acknowledgment. For example, “This work was supported in part by the U.S. Department of Commerce under Grant BS123456.”

The next few paragraphs should contain the authors’ current affiliations, including current address and e-mail. For example, F. A. Author is with the National Institute of Standards and Technology, Boulder, CO 80305 USA (e-mail:author@boulder.nist.gov).

S. B. Author, Jr., was with Rice University, Houston, TX 77005 USA. He is now with the Department of Physics, Colorado State University, Fort Collins, CO 80523 USA (e-mail: author@lamar.colostate.edu).

T. C. Author is with the Electrical Engineering Department, University of Colorado, Boulder, CO 80309 USA, on leave from the National Research Institute for Metals, Tsukuba, Japan (e-mail: author@nrim.go.jp).

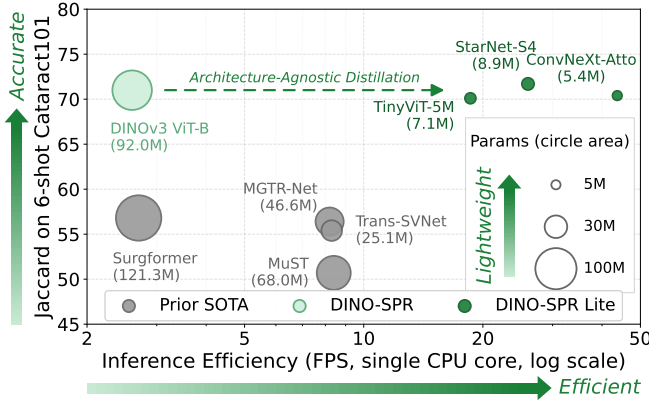


Fig. 1. 6-shot Cataract101 Jaccard against inference efficiency on CPU, with circle area proportional to parameter count. Prior methods (gray) stay below 8.4 FPS, while the SPR-Lite students (dark green) deliver accuracy comparable to the SPR-Teacher teacher (light green) at 18.6 to 43.7 FPS.

Although SPR-Teacher breaks the data reliance of the FM paradigm, it still inherits the second deficiency: as a heavy ViT, it incurs substantial computational overhead and rigid deployment constraints. To cure the real-time deficiency, we propose an architecture-agnostic distillation strategy that transfers the teacher into SPR-Lite. To bridge the teacher-student structural discrepancy, a cross-attention projection (CAP) maps heterogeneous student feature maps into the teacher's token space, enabling flexible supervision across CNN, ViT, and hybrid architectures. Complementing this alignment, a spatiotemporally-guided distillation (STGD) transfers spatial and temporal priors into intermediate student representations, supplying the dense supervisory signals missing under scarce annotations. As shown in Fig. 1, select SPR-Lite students drawn from three diverse backbone families exceed the 25 FPS real-time threshold on a single CPU core while approaching teacher-level accuracy, successfully reconciling accuracy and speed for clinical deployment.

Our contributions are summarized as follows:

- 1) We propose a redundancy-aware spatiotemporal adaptation framework for few-shot
- 2) SPR: SSTS compresses spatial redundancy via a soft top- K saliency gate and KPTF compresses temporal redundancy via keyframe-prior fusion, adapting a frozen image-level foundation model to surgical video with only a handful of labeled videos.
- 3) We propose an architecture-agnostic distillation scheme that compresses the adapted model into lightweight students of arbitrary backbone families: CAP projects heterogeneous feature maps into the teacher's token space, and STGD transfers the teacher's where- and when-to-look priors as dense supervision to compensate for scarce labels, enabling real-time CPU inference.
- 4) Extensive experiments on four public and one private datasets show that our method surpasses prior state-of-the-art by large margins under few-shot settings, while the distilled Lite models approach this accuracy with under 9M parameters at over 25 FPS on a single CPU core.

II. RELATED WORK

A. Surgical Phase Recognition

Early surgical phase recognition relied on handcrafted endoscopic features paired with statistical classifiers such as SVMs [7] and HMMs [8], which exhibited limited capacity in modeling complex surgical dynamics. The advent of EndoNet [9] revolutionized this task by introducing automatic CNN feature learning. Subsequent frameworks integrated 2D CNNs with recurrent modules or LSTMs to capture temporal dependencies [10]–[12], while temporal convolutions [13] and 3D CNNs [14], [15] further unified spatial and temporal representations. Recently, transformer architectures have shifted the design toward self-attention mechanisms for sequence-level dynamics. These include Trans-SVNet [16], which merges spatial and temporal embeddings, video transformers like SKiT [17] and LoViT [18] combining feature extractors with temporal heads, and Surgformer [19] with divided spatiotemporal attention. Despite their strong performance, these transformer-based models remain constrained by heavy reliance on extensive annotated data and prohibitive computational costs, hampering practical clinical deployment.

B. Generic and Surgical Foundation Models

Beyond task-specific designs, FMs pre-trained on massive datasets acquire universal visual priors with strong transferability. Generic FMs span self-supervised learning (SSL) and contrastive learning paradigms across images and videos. In the image domain, DINOv3 [20] leverages self-distillation on 1.7 billion curated images to extract dense visual representations, while CLIP [21] aligns 400 million image-text pairs via contrastive learning. Video counterparts like VideoMAE2 [22], V-JEPA2 [23], and InternVideo2 [24] extend these concepts to temporal dynamics through masked video modeling and spatiotemporal alignment, exhibiting robust few-shot generalization. Parallely, surgical FMs adapt these pre-training strategies to medical domains. Image-level SSL models such as EndoViT [4], EndoDINO [25], SurgeNetXL [26], SurgFM [27], and EndoSSL [28] apply masked modeling or self-distillation to endoscopic frames, while video-level SSL models like Endo-FM [29], EndoFM-LV [30], SurgVISTA [5], and SurgMotion [6] pre-train on video clips. Additionally, surgical vision-language models such as SurgVLP [31], HecVL [32], PeskaVLP [33], and OphCLIP [34] align clinical video with textual domain knowledge. Despite their potential, surgical FMs remain sensitive to procedure distributions in pre-training data, while privacy restrictions often hinder open releases, restricting broader research usage. Yet whether generic FMs can substitute for surgical FMs in few-shot SPR remains under-explored, and even the strongest generic FM inherits the heavy architectures that preclude real-time deployment; we address both gaps in this work.

C. Few-Shot Adaptation and Token Reduction

Few-shot learning targets recognition from a handful of labeled examples and has been extensively studied for video

action recognition [35]. In medical imaging, where expert annotation is particularly costly, recent studies show that general visual priors transfer surprisingly well under minimal supervision: multimodal large language models with in-context learning match specialist networks on cancer pathology classification from a few examples [36], and large-scale evaluations of parameter-efficient fine-tuning (PEFT) report the largest gains precisely in the low-data regimes typical of clinical imaging [37]. Among PEFT methods, low-rank adaptation (LoRA) [38] is the most widely adopted, injecting small trainable rank-limited matrices while keeping the pretrained backbone frozen, thereby preserving general semantics and resisting overfitting on small datasets. Two further lines of work are closely related to ours. Token reduction keeps the most task-relevant tokens to accelerate ViTs [39]; rather than discarding tokens for efficiency, our SSTS reweights them under a saliency budget so that the spatial redundancy of surgical frames becomes an inductive bias for few-shot adaptation. Likewise, dynamic video prompting highlights frames with high temporal variance as keyframes [40]; our KPTF transfers this idea from prompt insertion to temporal pooling. Crucially, the selected tokens and keyframe weights are not by-products but become dense supervision signals in our distillation stage.

D. Knowledge Distillation for Lightweight Models

Knowledge distillation (KD) [41] compresses high-capacity models into lightweight students by transferring soft logits or intermediate representations. In practical scenarios, modern KD extends beyond standard output imitation to address diverse domain-specific challenges. For instance, methods like VFM-KT [42] and CustomKD [43] demonstrate that task-specific adaptation of foundation models before distillation yields stronger supervisory signals than direct transfer from generic teachers. Regarding architectural heterogeneity, bridging ViT teachers with CNN or hybrid students relies on cross-architecture feature alignment, token-space projections [44], or sequence mixing alignment [45]. In temporal contexts, distilling video architectures demands transferring spatiotemporal structures rather than frame-level features alone [46]. Similarly, under label scarcity, distillation becomes essential to compensate for sparse task supervision [47]. Together, these developments highlight that practical distillation often requires addressing model adaptation, architectural alignment, temporal modeling, and label efficiency in a unified framework.

III. METHOD

A. Overview

To overcome label scarcity in few-shot surgical phase recognition, we introduce **SPR-TEACHER**, which adapts the generalization of a generic visual FM to the surgical domain. As illustrated in Fig. 2, we first devise a spatiotemporal adaptation tuning that elevates independent frame-level features into sequence representations via SSTS and KPTF modules. By explicitly filtering the inherent redundancy of surgical videos to aggregate only the most discriminative spatiotemporal signals, this design imposes a strong inductive bias crucial for few-shot learning. For resource-constrained clinical deployment,

we further distill it, through architecture-agnostic distillation, into **SPR-LITE**, where CAP bridges representational gaps and STGD enriches sparse supervision with the teacher's priors.

B. Spatiotemporal Adaptation Tuning

Given a surgical video clip $\mathbf{X} \in \mathbb{R}^{T \times 3 \times H \times W}$ of T consecutive frames, our goal is to predict the surgical phase of the last frame. To leverage the robust representation of FMs, we employ DINOv3 as the frame-level feature extractor. While we instantiate the framework with DINOv3, the adaptation operators below only assume a frozen image-level FM that emits per-frame tokens, so the framework does not hinge on a specific backbone. Specifically, for each frame $\mathbf{X}_t \in \mathbb{R}^{3 \times H \times W}$, DINOv3 splits the image into a regular grid of N non-overlapping patches. Following linear projection and the prepending of a learnable class token $\mathbf{z}_{cls}$, the input sequence $\mathbf{z}^{(0)}$ is formulated as

$$\mathbf{z}^{(0)} = [\mathbf{z}_{cls}; \mathbf{E}\mathbf{x}_1; \dots; \mathbf{E}\mathbf{x}_N] + \mathbf{E}_{pos} \in \mathbb{R}^{(N+1) \times D}, \quad (1)$$

where $\mathbf{E}$ is the projection matrix, $\mathbf{x}_i$ is the flattened i -th patch, and $\mathbf{E}_{pos}$ is the position embedding. After propagating through L ViT blocks, DINOv3 outputs from its last layer:

$$\mathbf{c}_t = \mathbf{z}_t^{(L)}[0], \quad \mathbf{P}_t = \mathbf{z}_t^{(L)}[1:N]. \quad (2)$$

where class token $\mathbf{c}_t$ summarizes the global context and spatial tokens $\mathbf{P}_t$ retain its spatial details for the frame. Consequently, the video clip $\mathbf{X}$ is initially encoded into a sequence of **frame-independent token sets** $(\mathbf{c}_{1:T}, \mathbf{P}_{1:T})$ lacking temporal interactions. To establish spatiotemporal modeling, our adaptation progressively condenses these tokens into a compact representation. Specifically, SSTS integrates spatial details $\mathbf{P}_t$ into $\mathbf{c}_t$ to produce **saliency-condensed frame tokens** $\tilde{\mathbf{c}}_{1:T} \in \mathbb{R}^{T \times D}$, which KPTF subsequently aggregates across time into a unified **clip-level feature** $\tilde{\mathbf{h}} \in \mathbb{R}^D$ for phase classification.

1) Salient Spatial Token Selection: Rather than treating all patch tokens equally, SSTS condenses the spatial tokens of a frame into its frame token through a learned saliency gate that reinforces the informative regions. Saliency in a surgical frame is spatially contiguous. An instrument or a tissue spans a region rather than a single isolated patch. Each patch should therefore be rated partly by its surroundings, and since a patch is only meaningful in the frame's context, partly by the class token as well. We restore the patch tokens to the $\sqrt{N} \times \sqrt{N}$ grid and score each on its own and with the class token,

$$\hat{\mathbf{P}}_t = \mathbf{P}_t + f_{dw}(\mathbf{P}_t), \quad \tilde{\omega}_t = f_{loc}(\hat{\mathbf{P}}_t) + f_{glob}([\hat{\mathbf{P}}_t; \mathbf{c}_t]). \quad (3)$$

where f_{dw} is a depthwise 3×3 convolution over the reshaped patch grid, and f_{loc} and f_{glob} are patch-shared MLPs. The operator $[\cdot; \cdot]$ puts the class token next to every patch, and $\mathbf{p}_{t,i}$ and $\hat{\mathbf{p}}_{t,i}$ denote the i -th token of $\mathbf{P}_t$ and $\hat{\mathbf{P}}_t$. The score $\tilde{\omega}_t \in \mathbb{R}^N$ holds one entry per patch, and the refinement touches the patch tokens only, leaving the class token as the backbone produced it. A sigmoid turns the scores into per-patch weights, which project the patches through a shared linear $\mathbf{W}_v$ and pool

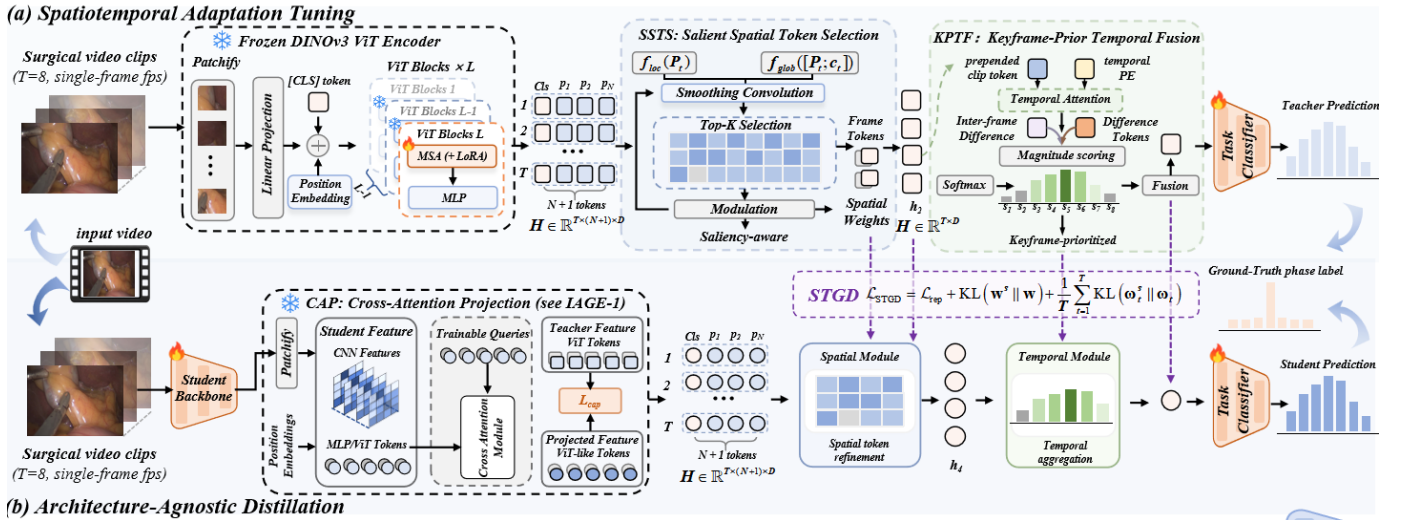


Fig. 2. Overview of the proposed framework. A frozen DINOv3 backbone is adapted to surgical video through a Salient Spatial Token Selection (SSTS) module and a Keyframe-Prior Temporal Fusion (KPTF) module, yielding SPR-Teacher. SPR-Lite further compresses the adapted model into lightweight backbones of arbitrary architectures via architecture-agnostic distillation.

them into the frame token,

$$\omega_t = \sigma(\tilde{\omega}_t), \quad \tilde{\mathbf{c}}_t = \mathbf{c}_t + \alpha \frac{\sum_i \omega_{t,i} \mathbf{W}_v \mathbf{p}_{t,i}}{\sum_i \omega_{t,i}}. \quad (4)$$

With α a learnable scalar, the gate $\omega_t \in (0, 1)^N$ a per-patch weight, and $\tilde{\mathbf{c}}_t \in \mathbb{R}^D$ the saliency-condensed frame token.

In a surgical video the background occupies most of the field and the instrument or the tissue that matters is a small share of the patches, so the gate should be sharpened to concentrate the frame token on that share. To this end we add a single training-time term built on a soft top- K selection of the saliency scores, which drives the gate up on the few patches that carry the content of the frame and down on the background that does not. Let $\mathcal{S}_t$ hold the indices of the K largest entries of $\tilde{\omega}_t$, and the term be the cross-entropy of the gate against that selection,

$$L_{\text{top-}K} = -\frac{1}{N} \left[\sum_{i \in \mathcal{S}_t} \log \omega_{t,i} + \sum_{i \notin \mathcal{S}_t} \log(1 - \omega_{t,i}) \right]. \quad (5)$$

The term pulls the gate towards the top- K selection, and it is smallest once the two agree. Since a patch that stays in between is penalized whichever side of $\mathcal{S}_t$ it falls on, agreement can only be reached by saturation, which sends the background towards 0 and the selected patches towards 1 and turns the graded field of Eq. (4) into a soft top- K . SSTS scores each patch with its spatial neighborhood and the frame context, suppresses the background through a soft top- K , and fuses the patches into the frame token.

2) Keyframe-Prior Temporal Fusion: The saliency-condensed frame tokens stay isolated between frames, so KPTF carries out the temporal modeling that fuses the sequence into a clip-level feature. A learnable clip token $\mathbf{h}$ is prepended to the sequence and lets every frame gather cross-frame context through temporal attention, whose output at the clip position is the clip summary $\hat{\mathbf{h}}$,

$$\hat{\mathbf{h}} = \text{TAttn}([\mathbf{h}, \tilde{\mathbf{c}}_1 + \mathbf{e}_1, \tilde{\mathbf{c}}_2 + \mathbf{e}_2, \dots, \tilde{\mathbf{c}}_T + \mathbf{e}_T])[0]. \quad (6)$$

where TAttn is a transformer block over the $T + 1$ tokens of the augmented sequence, and $\mathbf{e}_t$ a learnable temporal positional encoding that keeps the frame order.

Surgical video is temporally redundant, as the field holds a steady state through most of a phase, and the frames that differ markedly from their neighbors carry the discriminative evidence. The keyframe-scoring branch, inspired by inter-frame temporal prompting [40], rates each frame by its inter-frame difference $\delta_t = \tilde{\mathbf{c}}_t - \tilde{\mathbf{c}}_{t-1}$ (with $\delta_1 = \mathbf{0}$), so that a frame of marked change outranks one that holds steady. A one-dimensional depthwise convolution $f_{\text{dw-1}}$ of kernel size 3 aggregates the difference sequence, whose mean squared magnitude gives the per-frame score $\tilde{\mathbf{w}}_t$. Softmax turns the scores into the frame weights $\mathbf{w}$,

$$\tilde{\mathbf{w}}_t = \frac{1}{D} \|f_{\text{dw-1}}(\delta_{1:T})_t\|^2, \quad \mathbf{w} = \text{softmax}(\tilde{\mathbf{w}}_{1:T}). \quad (7)$$

The weights quantify how much each frame contributes to the clip and pool the sequence into a keyframe-pooled token $\mathbf{v} = \sum_t w_t \tilde{\mathbf{c}}_t$. The clip summary and the keyframe-pooled token carry complementary evidence, the context of the whole clip and the motion of its most informative frames, so a fusion operator f_{fuse} merges them,

$$\hat{\mathbf{h}} = f_{\text{fuse}}([\hat{\mathbf{h}}; \mathbf{v}]). \quad (8)$$

KPTF fuses the clip summary and the keyframes into the clip-level feature $\hat{\mathbf{h}}$, which captures the informative motions and yields a more accurate phase prediction.

3) Adaptation Objective: The DINOv3 backbone stays largely frozen to preserve its general pretrained semantics. Only a small set of parameters is trained, the LoRA adapters [38] on the query/value projections of the last ViT block, the two adaptation modules SSTS and KPTF, and a linear classification head f_{cls} . Training minimizes the phase cross-entropy together with the top- K selection term of SSTS,

$$\hat{\mathbf{y}} = f_{\text{cls}}(\hat{\mathbf{h}}), \quad L_{\text{tuning}} = L_{\text{CE}} + L_{\text{top-}K}. \quad (9)$$

C. Architecture-Agnostic Distillation

SPR-Teacher inherits the generalization of DINOv3 and delivers promising few-shot performance, but it remains tied to the heavy ViT architecture, which restricts deployment flexibility and slows both loading and inference. We therefore distill it into **SPR-LITE**, whose backbone can follow any mainstream lightweight architecture. Those architectures fall into three families: CNN, ViT, and hybrid. A ViT student already emits a class token and patch tokens in the teacher's form, whereas the other two architectures emit feature maps whose spatial cells might not match the teacher's tokens, so the teacher's spatial and temporal priors cannot be applied to them directly. CAP resolves this at the last stage of the student, transforming the feature map into the teacher's token space so that the student carries the same per-frame tokens as the teacher. The student then employs SSTS and KPTF of the same structure, which is what makes the scheme architecture-agnostic. STGD compensates for the scarce supervision of the few-shot regime with the teacher's spatiotemporal priors.

1) Student Architecture: We take CNN as the representative student. It is image-based and exposes a last-stage feature map $\mathbf{M}_t \in \mathbb{R}^{C \times h \times w}$, which CAP maps into the teacher's token space, producing a class token $\mathbf{c}_t^s$ and N patch tokens $\mathbf{P}_t^s$ of width D . The student therefore yields tokens of the same form as the teacher, which its SSTS and KPTF aggregate progressively into the saliency-condensed frame tokens $\tilde{\mathbf{c}}_{1:T}^s$ and then the clip-level feature $\tilde{\mathbf{h}}^s$, from which the classification head gives the clip prediction $\hat{\mathbf{y}}^s$.

2) Distillation Scheme: Cross-Attention Projection (CAP). CAP is the front end of the student pipeline, converting the feature map into the token form that every later module takes as input. Its design follows ScaleKD [44] and is built as a cross-attention interface. The last stage holds the most task-specific representation, so CAP converts the feature map there. It flattens $\mathbf{M}_t$ into $h \times w$ spatial tokens and maps each token from C to the teacher's width D with a shared linear projection, adding positional embeddings that keep the grid layout. Each student token then has the teacher's width, so only their number remains to reconcile. CAP resolves the count mismatch on the query side. It keeps a set of learnable queries in the teacher's space, one **per teacher token:** a patch query for each of the teacher's patches, plus a single class query for the class slot the student lacks. Cross-attention emits one output per query, so the output length is decided by the queries rather than by the feature map. The $h \times w$ cells are therefore aggregated into exactly as many patch tokens $\mathbf{P}_t^s$ as the teacher has. Each patch query carries the position of the teacher patch it stands for, so a query draws most strongly on the student cells occupying the same region of the frame. The student's patch tokens thus keep a spatial layout, which the grid operations of SSTS rely on, and the class query produces the student's class token $\mathbf{c}_t^s$. A squared ℓ_2 distance then aligns the student tokens with the teacher's,

$$L_{\text{cap}} = \frac{1}{T} \sum_{t=1}^T \left\| [\mathbf{c}_t; \mathbf{P}_t] - \text{CAP}(\mathbf{M}_t; \mathbf{q}_{\text{cls}}, \mathbf{q}_{1:N}) \right\|_2^2. \quad (10)$$

Spatiotemporally-Guided Distillation (STGD). To address

TABLE I

SUMMARY OF THE FIVE SURGICAL PHASE RECOGNITION DATASETS USED IN THIS STUDY.

Dataset	Abbrev.	Country	#Videos (train/val/test)	#Phases	Duration
Cholec80	C80	France	32/8/40	7	51.3 h
M2CAI16	M16	France	22/5/14	8	26.3 h
Cataract101	Cat101	Germany	60/15/26	10	14.0 h
CatPriv16	CP16	China	25/5/11	16	10.0 h
PitVis	PV	UK	19/5/5	14	32.1 h

the sparsity of few-shot supervision, STGD distills the teacher's spatiotemporal knowledge through four intermediate signals. The representation signals are the saliency-condensed frame tokens $\tilde{\mathbf{c}}_{1:T} \in \mathbb{R}^{T \times D}$ and the clip-level feature $\tilde{\mathbf{h}} \in \mathbb{R}^D$, while the prior signals are the spatial gates $\omega_{1:T} \in \mathbb{R}^{T \times N}$ and the temporal weights $\mathbf{w} \in \mathbb{R}^T$. **The temporal weights $\mathbf{w}$ form a distribution over frames, and the spatial gates ω_t are normalized over the patches,** so each enters the KL as a distribution. The prior signals encode where and when the teacher attends, information that the scarce labels do not provide. The student pipeline exposes the same four signals $\tilde{\mathbf{c}}_{1:T}^s$, $\tilde{\mathbf{h}}^s$, $\omega_{1:T}^s$, and $\mathbf{w}^s$, and aligns each to the teacher's counterpart. We group the representation terms into L_{rep} and add the prior terms,

$$L_{\text{rep}} = d_{\cos}(\tilde{\mathbf{h}}^s, \tilde{\mathbf{h}}) + \frac{1}{T} \sum_{t=1}^T d_{\cos}(\tilde{\mathbf{c}}_t^s, \tilde{\mathbf{c}}_t) \quad (11)$$

$$L_{\text{stgd}} = L_{\text{rep}} + \text{KL}(\mathbf{w}^s \parallel \mathbf{w}) + \frac{1}{T} \sum_{t=1}^T \text{KL}(\omega_t^s \parallel \omega_t)$$

where $d_{\cos}$ is the cosine distance. Together with L_{cap} , these terms supervise every student module, supplementing the sparse few-shot labels with a dense distillation signal.

Overall objective. The student is trained with the **teacher's tuning objective plus the** two distillation terms,

$$L = L_{\text{tuning}} + L_{\text{cap}} + L_{\text{stgd}}. \quad (12)$$

IV. EXPERIMENTS

A. Implementation Details

We adopt DINOv3 ViT-B as the default backbone with $T=8$ clips at 224×224 . Each frame is split into $N=196$ patch tokens of dimension $D=768$, and the budget of the SSTS selection term is set to $K=0.3N$. All models use AdamW (weight decay 0.05) with cosine schedule and 5-epoch warmup, gradient accumulation of 4 at batch size 12, label smoothing 0.1, Mixup and CutMix, and EMA (decay 0.999). For **SPR-Teacher**, we keep the backbone largely frozen, training only LoRA adapters of the last layer, SSTS, KPTF, and the linear classification head for 80 epochs ($1r \ 5 \times 10^{-4}$). For **SPR-Lite**, a lightweight student **is supervised by the frozen SPR-Teacher teacher and trained for 100 epochs.** All training runs on a single RTX 4090 (24 GB) with PyTorch 2.6 and CUDA 12.4.

B. Evaluation

1) Datasets: To evaluate across diverse surgical scenarios, we curate five SPR datasets spanning three procedure types:

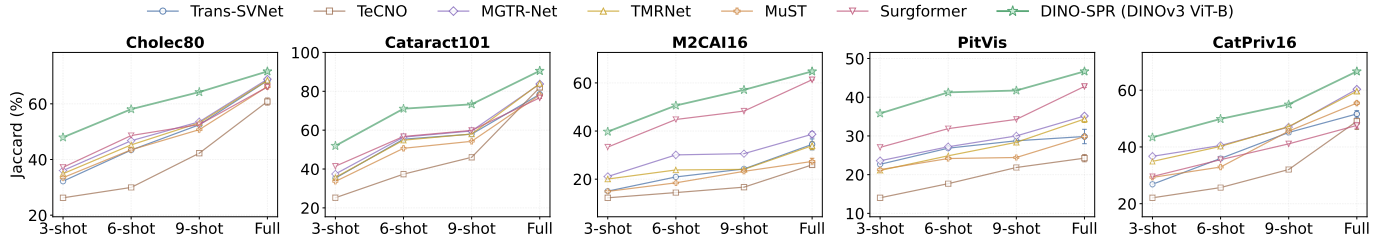


Fig. 3. Few-shot trends (3-/6-/9-shot) of SPR-Teacher and prior methods on Jaccard, averaged over five datasets.

laparoscopic cholecystectomy (Cholec80, M2CAI16), cataract surgery (Cataract101, CatPriv16), and transsphenoidal pituitary surgery (PitVis), as summarized in Table I. Among them, CatPriv16 is our private dataset providing an unseen reference for surgical FMs. All videos are sampled at 1 fps, and few-shot splits use 3, 6, or 9 labeled training videos with unchanged validation and test sets.

2) *Metrics*: Following prior work [11], [19], we report accuracy, precision, recall, F1-score, and Jaccard under unrelaxed evaluation, i.e., strict frame-by-frame scoring without temporal smoothing, averaged over three random seeds. We prioritize Jaccard for jointly penalizing frame-level false positives and negatives, and summarize cross-dataset results by the mean Jaccard together with the range $\Delta = \max - \min$, where a smaller Δ indicates more balanced generalization. For deployment efficiency, we additionally report the parameter count, per-frame GFLOPs, and the FPS of a single $T=8$ clip measured on a single core of an AMD EPYC 7543 CPU.

C. Comparisons with SOTAs

1) *Few-shot Comparison with SCTAs*: We compare SPR-Teacher with eight prior methods spanning two paradigms. Multi-stage methods, namely Trans-SVNet [16], TeCNO [13], MGTR-Net [48], TMRNet [11], MuST [49], LoViT [18], and SKiT [17], fine-tune a backbone before appending a temporal model. Surgformer [19], the only end-to-end approach, instead jointly models multiple input frames through spatiotemporal attention in a ViT. Tables III and II report the 6-shot comparison over the four public datasets and our private CatPriv16, respectively.

Under few-shot supervision, multi-stage methods generally lag behind, since scarce labels do not allow their feature

extractors to be fine-tuned well, whereas the end-to-end Surgformer performs somewhat better. With the generic FM prior and our redundancy-aware spatiotemporal adaptation, SPR-Teacher clearly surpasses all prior methods on every metric and dataset, reaching 58.1, 71.0, and 49.9 Jaccard on Cholec80, Cataract101, and CatPriv16. These scores improve on the strongest prior method by 9.5, 14.2, and 9.4 on each dataset, and the gains span 5.8 to 14.2 across the five datasets. Figure 3 further shows that SPR-Teacher leads by a large margin at every label budget. The advantage is largest at the extreme 3-shot setting, for instance on M2CAI16, where SPR-Teacher reaches 39.7 Jaccard while the prior method TeCNO falls to 12.3, demonstrating the robustness of our method to extreme label scarcity.

2) *Computational Cost and Parameter Efficiency*: Clinical deployment often has to work without a dedicated GPU, so we emulate such a limited environment by measuring all inference on a single CPU core, as reported in Table IV for 6-shot Cholec80. Prior methods run at only 2.7 to 8.4 FPS, far below the 25 FPS native video rate. Surgformer is the slowest because it jointly attends over all frames of a clip with a 121M ViT. SPR-Teacher reaches 58.1 Jaccard, yet as a ViT it inherits the same limit and runs at 2.6 FPS. High performance alone thus does not make a model deployable. Distilling SPR-Teacher into lightweight students in an architecture-agnostic manner closes that gap and meets the real-time requirement, reaching up to 44 FPS on a single CPU with performance close to SPR-Teacher.

D. Ablation Study

1) *Generic versus Surgical Foundation Models*: To justify the choice of the generic visual prior, we freeze each FM as a feature extractor and train MGTR-Net [48] as the temporal model on 8-frame windows, under the unified 6-shot protocol across five datasets; the FM configurations are detailed in Table IX (Appendix I). Since surgical FMs may have seen the evaluation datasets during pretraining, we highlight our private CatPriv16 as a leakage-free reference for fair generalization.

Figure 4 ranks all FMs by mean Jaccard. Generic FMs achieve higher Jaccard with more stable cross-dataset generalization than surgical FMs. On the leakage-free CatPriv16, surgical FMs nearly all rank at their worst, suggesting that their public-dataset scores overestimate generalization to unseen procedures. Among generic FMs, self-supervised models rank ahead of contrastive ones in this evaluation. Video-level input also brings no clear gain over image-level input at

TABLE II
COMPARISON OF METHODS ON 6-SHOT SURGICAL PHASE
RECOGNITION ON OUR PRIVATE CATPRIV16 DATASET.

Method	Accuracy	Precision	Recall	F1	Jaccard
Trans-SVNet	68.2±0.4	55.3±1.0	45.1±1.7	45.0±1.0	35.9±0.9
TeCNO	56.4±2.0	39.8±3.1	36.8±3.7	32.5±3.1	25.7±2.7
MGTR-Net	70.3±0.7	54.2±1.2	50.8±0.7	48.6±0.8	40.5±0.7
TMRNet	70.4±0.6	54.5±0.8	50.6±0.5	48.6±0.7	40.2±0.7
Surgformer	66.3±0.4	49.4±0.1	44.7±1.4	43.1±1.2	35.5±1.0
MuST	65.7±0.2	49.6±1.4	43.5±0.2	42.1±0.4	32.9±0.3
LoViT [†]	59.2±2.0	36.5±0.4	35.7±0.4	31.6±0.4	24.0±0.3
SKiT [†]	62.4±0.8	36.5±0.7	36.8±1.0	33.2±0.1	26.2±0.2
SPR-Teacher	74.5±0.2	71.8±1.9	60.1±1.2	58.9±0.7	49.9±0.3

TABLE III

COMPARISON OF METHODS ON 6-SHOT SURGICAL PHASE RECOGNITION ACROSS FOUR PUBLIC DATASETS. BEST RESULTS ARE IN **BOLD**. RESULTS REPORT MEAN $\pm$ STD ACROSS THREE INDEPENDENT RUNS UNDER UNRELAXED EVALUATION.

Method	Paradigm	Accuracy	Precision	Recall	F1	Jaccard	Accuracy	Precision	Recall	F1	Jaccard
Trans-SVNet	Multi-stage	72.8 $\pm$ 0.5	62.8 $\pm$ 1.9	61.2 $\pm$ 0.4	55.9 $\pm$ 1.0	43.4 $\pm$ 0.9	74.8 $\pm$ 1.5	72.3 $\pm$ 1.2	70.3 $\pm$ 1.6	67.1 $\pm$ 1.6	55.4 $\pm$ 2.0
TeCNO	Multi-stage	56.6 $\pm$ 3.7	46.8 $\pm$ 1.8	49.5 $\pm$ 0.5	41.3 $\pm$ 1.7	30.0 $\pm$ 1.6	58.9 $\pm$ 1.5	53.0 $\pm$ 1.7	54.0 $\pm$ 1.5	48.6 $\pm$ 1.3	37.4 $\pm$ 1.1
MGTR-Net	Multi-stage	73.2 $\pm$ 0.4	65.9 $\pm$ 0.6	64.6 $\pm$ 0.6	59.3 $\pm$ 0.6	46.7 $\pm$ 0.6	73.3 $\pm$ 0.4	73.6 $\pm$ 0.6	70.8 $\pm$ 0.5	66.8 $\pm$ 0.4	56.4 $\pm$ 0.5
TMRNet	Multi-stage	72.1 $\pm$ 0.3	63.6 $\pm$ 0.3	63.6 $\pm$ 0.6	58.0 $\pm$ 0.4	45.2 $\pm$ 0.4	72.6 $\pm$ 0.5	71.7 $\pm$ 0.7	69.8 $\pm$ 1.0	65.3 $\pm$ 0.5	54.8 $\pm$ 0.6
Surgformer	End-to-end	64.4 $\pm$ 0.9	59.7 $\pm$ 0.1	62.8 $\pm$ 0.6	58.2 $\pm$ 0.4	48.6 $\pm$ 0.5	72.4 $\pm$ 0.3	69.9 $\pm$ 0.5	70.1 $\pm$ 0.2	66.9 $\pm$ 0.3	56.8 $\pm$ 0.3
MuST	Multi-stage	70.3 $\pm$ 0.5	60.2 $\pm$ 0.8	64.1 $\pm$ 0.8	57.0 $\pm$ 0.9	43.5 $\pm$ 0.8	70.2 $\pm$ 0.5	66.8 $\pm$ 0.5	68.3 $\pm$ 0.5	63.6 $\pm$ 0.4	50.7 $\pm$ 0.5
LoViT [†]	Multi-stage	61.7 $\pm$ 0.2	47.3 $\pm$ 3.4	51.0 $\pm$ 2.5	43.3 $\pm$ 2.8	31.4 $\pm$ 1.8	34.1 $\pm$ 8.4	55.4 $\pm$ 5.6	32.3 $\pm$ 7.0	28.5 $\pm$ 7.4	19.8 $\pm$ 5.7
SKiT [†]	Multi-stage	63.3 $\pm$ 1.2	47.3 $\pm$ 4.8	48.9 $\pm$ 3.2	42.7 $\pm$ 2.6	31.4 $\pm$ 1.9	39.2 $\pm$ 8.4	59.4 $\pm$ 5.7	35.8 $\pm$ 8.4	33.5 $\pm$ 9.0	24.0 $\pm$ 7.2
SPR-Teacher (DINOv3 ViT-B)	End-to-end	80.1$\pm$0.2	72.7$\pm$0.1	76.6$\pm$0.8	70.7$\pm$0.5	58.1$\pm$0.4	85.2$\pm$0.1	83.4$\pm$0.1	82.7$\pm$0.1	79.5$\pm$0.2	71.0$\pm$0.2
Trans-SVNet	Multi-stage	45.3 $\pm$ 3.7	30.6 $\pm$ 3.4	32.5 $\pm$ 2.7	27.6 $\pm$ 3.4	20.9 $\pm$ 2.5	58.7 $\pm$ 0.8	42.2 $\pm$ 2.2	38.7 $\pm$ 0.6	35.9 $\pm$ 0.8	26.8 $\pm$ 0.6
TeCNO	Multi-stage	31.1 $\pm$ 3.4	26.3 $\pm$ 1.8	26.5 $\pm$ 0.3	22.2 $\pm$ 0.7	14.4 $\pm$ 0.6	44.6 $\pm$ 2.9	29.0 $\pm$ 1.3	27.7 $\pm$ 3.4	24.7 $\pm$ 1.9	17.7 $\pm$ 1.5
MGTR-Net	Multi-stage	44.4 $\pm$ 1.8	44.5 $\pm$ 1.0	44.3 $\pm$ 0.6	38.9 $\pm$ 0.4	30.1 $\pm$ 0.2	56.7 $\pm$ 0.2	40.6 $\pm$ 0.8	41.9 $\pm$ 0.2	36.6 $\pm$ 0.1	27.2 $\pm$ 0.1
TMRNet	Multi-stage	38.3 $\pm$ 4.8	37.3 $\pm$ 0.4	38.8 $\pm$ 2.9	32.3 $\pm$ 2.4	23.8 $\pm$ 1.5	55.3 $\pm$ 1.3	38.5 $\pm$ 1.5	40.4 $\pm$ 0.7	34.3 $\pm$ 0.9	24.9 $\pm$ 0.8
Surgformer	End-to-end	61.6 $\pm$ 0.5	55.3 $\pm$ 0.5	59.9 $\pm$ 0.7	54.0 $\pm$ 0.6	44.8 $\pm$ 0.8	52.1 $\pm$ 0.7	40.4 $\pm$ 0.7	38.0 $\pm$ 0.5	35.8 $\pm$ 0.6	27.1 $\pm$ 0.5
MuST	Multi-stage	39.0 $\pm$ 0.3	30.5 $\pm$ 1.0	34.5 $\pm$ 0.2	27.0 $\pm$ 0.5	18.5 $\pm$ 0.3	53.5 $\pm$ 0.4	40.0 $\pm$ 1.3	37.9 $\pm$ 0.4	33.7 $\pm$ 0.3	24.2 $\pm$ 0.3
LoViT [†]	Multi-stage	28.0 $\pm$ 8.9	15.7 $\pm$ 2.6	17.2 $\pm$ 1.5	12.2 $\pm$ 2.0	7.6 $\pm$ 1.5	47.5 $\pm$ 5.0	26.1 $\pm$ 4.6	24.3 $\pm$ 2.6	20.7 $\pm$ 3.8	15.0 $\pm$ 3.1
SKiT [†]	Multi-stage	13.9 $\pm$ 4.0	13.9 $\pm$ 4.3	17.9 $\pm$ 1.3	9.2 $\pm$ 3.4	5.3 $\pm$ 2.0	47.8 $\pm$ 6.0	24.5 $\pm$ 5.4	24.4 $\pm$ 2.9	20.8 $\pm$ 4.2	14.5 $\pm$ 3.3
SPR-Teacher (DINOv3 ViT-B)	End-to-end	72.3$\pm$0.8	64.9$\pm$1.1	71.5$\pm$0.5	62.9$\pm$0.2	50.6$\pm$0.3	67.9$\pm$1.1	59.1$\pm$1.0	53.8$\pm$0.6	52.9$\pm$0.3	41.3$\pm$0.3

[†]LoViT and SKiT borrow the fine-tuned ResNet50 backbone from Trans-SVNet, as their original feature extractors are not publicly available.

TABLE IV

COMPUTATIONAL COST AND PERFORMANCE ON 6-SHOT CHOLEC80.

Method	Params $\downarrow$	GFLOPs $\downarrow$	FPS $\uparrow$	Jaccard $\uparrow$
Prior SOTA				
Trans-SVNet	25.1M	4.1	8.3	43.4
MGTR-Net	46.6M	4.1	8.2	46.7
MuST	68.0M	10.6	8.4	43.5
Surgformer	121.3M	22.5	2.7	48.6
SPR-Teacher				
DINOv3 ViT-B	92.0M	17.6	2.6	58.1
SPR-Lite				
TinyViT-5M	7.1M	1.2	18.6	55.4
StarNet-S4	8.9M	1.1	26.0	55.2
ConvNeXt-Atto	5.4M	0.6	43.7	52.2

a much larger computational cost: VideoMAE2 ViT-g with 1B parameters reaches 47.2, only marginally above the 86M DINOv3 ViT-B at 46.9. We therefore adopt DINOv3 [20] as the backbone for its strongest few-shot generalization.

2) *DINOv3 Pretraining and Scale*: To understand DINOv3's promising few-shot performance, we first probe the backbone with single frames ($T=1$), before adding any temporal module. Table V reports this ablation, from which three findings emerge. First, pretraining largely decides the outcome, with Jaccard climbing steadily from random to ImageNet-1K to DINOv3 weights even when the features are frozen. Fine-tuned DINOv3 ViT-B reaches 48.6 Jaccard on Cholec80 at 6-shot, versus 42.9 for ImageNet-1K. Second, scaling yields diminishing returns beyond ViT-B. Moving from ViT-S to ViT-B brings clear gains, whereas ViT-L improves only marginally at $3.5\times$ the parameters (304M versus 86M). Third, fine-tuning consistently outperforms frozen features. These findings lead us to adopt DINOv3 ViT-B for its accuracy-efficiency balance and to fine-tune it in all subsequent experiments.

3) *Spatiotemporal Modeling Design*: Table VI ablates spatial and temporal design choices on Cataract101 and Cholec80 at 6-shot, from single frames to full temporal fusion. In R1, we freeze the DINOv3 backbone and fine-tune a linear classifier on its CLS token, which yields only 34.6 and 38.8 Jaccard.

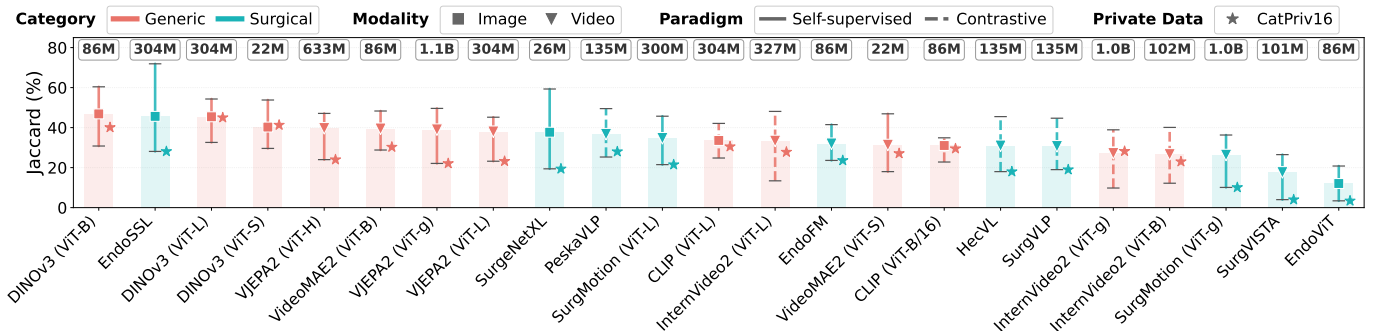


Fig. 4. Ablation of foundation models as the frozen backbone under 6-shot surgical phase recognition with MGTR-Net as the temporal model. For each foundation model, sorted by decreasing mean Jaccard, the vertical bar spans the min-max Jaccard across the five datasets, and the shape marker marks the mean Jaccard. The star marks the private CatPriv16 dataset as an unseen domain, and the labeled value is the parameter size.

TABLE V

BACKBONE PRETRAINING AND FINE-TUNING ABLATION AT T=1 (SINGLE FRAME). WE COMPARE RANDOM, IMAGENET-1K, AND DINOv3 WEIGHTS UNDER FROZEN AND FINE-TUNED MODES ACROSS ViT SIZES.

Weights	Mode	Cholec80			Cataract101		
		3-shot	6-shot	9-shot	3-shot	6-shot	9-shot
ViT-S (22M)							
Random	Finetune	21.2	24.8	28.0	10.9	16.0	16.7
IN-1K	Frozen	30.5	33.8	35.6	22.4	29.4	31.3
IN-1K	Finetune	37.8	42.9	51.7	40.2	55.8	58.4
DINOv3	Frozen	33.0	37.2	39.7	26.8	33.7	35.5
DINOv3	Finetune	37.4	44.7	51.9	34.7	55.4	55.1
ViT-B (86M)							
Random	Finetune	20.3	25.4	29.3	13.1	17.8	20.3
IN-1K	Frozen	34.0	37.5	39.7	25.7	35.2	37.2
IN-1K	Finetune	37.0	42.9	50.1	42.2	57.2	58.7
DINOv3	Frozen	35.6	38.8	42.3	26.1	34.6	35.3
DINOv3	Finetune	40.9	48.6	54.7	40.9	56.9	62.2
ViT-L (304M)							
Random	Finetune	21.0	26.2	31.3	14.5	19.9	22.0
IN-1K	Frozen	33.2	41.3	43.0	31.9	43.3	46.2
IN-1K	Finetune	37.2	44.4	49.4	45.9	56.5	56.7
DINOv3	Frozen	36.2	42.0	45.0	26.2	35.0	36.4
DINOv3	Finetune	42.7	49.7	55.3	47.3	61.8	63.0

TABLE VI

SPATIOTEMPORAL DESIGN ABLATION ON 6-SHOT TRAINING.

Method	T	Cataract101	Cholec80
R1. Single frame (T=1): spatial design			
Frozen DINOv3 (CLS)	1	34.6	38.8
Fine-tuned DINOv3 (full)	1	55.0 (+20.4)	46.1 (+7.3)
Frozen + LoRA (CLS)	1	55.2 (+20.6)	45.5 (+6.7)
Frozen + LoRA + mean-patch	1	53.0 (+18.4)	45.8 (+7.0)
Frozen + LoRA + SSTS (ours)	1	57.7 (+23.1)	49.6 (+10.8)
R2. Simple temporal fusion			
Frozen+LoRA+SSTS (T=1)	1	57.7	49.6
+ Mean Pooling	4	57.8 (+0.1)	49.8 (+0.2)
+ Mean Pooling	8	58.2 (+0.5)	51.2 (+1.6)
+ Max Pooling	4	61.5 (+3.8)	49.0 (-0.6)
+ Max Pooling	8	62.8 (+5.1)	51.5 (+1.9)
+ Bi-LSTM	4	60.2 (+2.5)	52.0 (+2.4)
+ Bi-LSTM	8	65.0 (+7.3)	53.5 (+3.9)
R3. Spatiotemporal fusion (ours)			
Frozen+LoRA+SSTS (T=1)	1	57.7	49.6
+ KPTF	4	64.6 (+6.9)	53.9 (+4.3)
+ KPTF	8	71.0 (+13.3)	58.1 (+8.5)
+ KPTF	16	76.1 (+18.4)	57.3 (+7.7)

LoRA fine-tuning, which **closely** matches full fine-tuning, lifts Jaccard to 55.2 **and 45.5**, and our SSTS module further raises it to 57.7 **and 49.6**, confirming the value of salient spatial token selection. In R2, temporal fusion over this R1-best setup adds clear gains as the clip lengthens, and a lightweight Bi-LSTM is the strongest simple aggregator, reaching 65.0 **and 53.5** at T=8. In R3, our KPTF module, which guides temporal fusion by keyframe priors, surpasses Bi-LSTM at every clip length, reaching 71.0 **and 58.1** at T=8. **As the gains do not grow consistently with longer clips while the latency increases**, we adopt T=8 as the **accuracy-efficiency balance**.

4) **Lightweight Backbone Choices for SPR-Lite**: Table VII compares each student trained from scratch and after

architecture-agnostic distillation on Cataract101 and Cholec80. Distillation improves every mainstream lightweight backbone across the three architecture categories, by over 20 Jaccard on average on Cataract101, confirming that it transfers **SPR-Teacher** knowledge to heterogeneous students. In particular, the strongest student of each family reaches 72.2, 72.3, and 71.0 for ConvNeXt-Tiny, TinyViT-11M, and MobileViTv2-150, respectively, matching or exceeding the 92M teacher's 71.0 on Cataract101. Distillation thus allows picking the backbone best suited to the deployment setting with little performance loss.

5) **Distillation Component Ablation**: We ablate the distillation components on the heterogeneous CNN ConvNeXt-Atto in Table VIII. Training **the student** with the task loss alone stays weak at 50.2 on Cataract101, because the scarce labels offer little supervision to the small model. **Each of L_{cap} and L_{stgd} alone improves over this baseline, to 62.4 and 59.0 on Cataract101, and combining the two yields the largest gain, reaching 71.6**; Cholec80 follows the same trend, rising to 52.2 from the 39.8 baseline. **The two terms are thus complementary in the few-shot regime: representation alignment and spatiotemporal prior transfer address different gaps left by the scarce labels.**

E. Qualitative Analysis

1) **Frame-level Phase Recognition**: Figure 5 compares per-frame predictions of **SPR-Teacher** with prior methods on test videos from Cholec80 and Cataract101. **SPR-Teacher** produces more continuous and accurate phase recognition, with cleaner boundaries and far fewer fragmented predictions. The improvement is most pronounced at phase transitions, where the spatiotemporal module captures discriminative cues across frames to keep the prediction coherent, while prior methods oscillate rapidly between adjacent phases. Overall, the predicted sequences align closely with the ground-truth an-

TABLE VII

DISTILLATION OF **SPR-TEACHER** INTO LIGHTWEIGHT STUDENTS ON 6-SHOT SETTINGS. EACH CELL: STUDENT-ONLY → DISTILLED.

Model	Params↓	Cataract101↑	Cholec80↑
Teacher			
SPR-Teacher (ViT-B)	92M	71.0	58.1
Pure CNN			
ConvNeXt-Atto	5.4M	50.2 → 71.6	39.8 → 52.2
ConvNeXt-Tiny	33.3M	53.6 → 72.2	43.3 → 57.8
StarNet-S2	5.1M	51.3 → 69.7	36.6 → 52.3
StarNet-S4	8.9M	52.4 → 71.7	44.2 → 55.2
MobileNetV3-Small	3.6M	50.2 → 66.8	43.1 → 53.0
ViT-based			
TinyViT-5M	7.1M	45.8 → 70.9	41.9 → 55.4
TinyViT-11M	12.8M	47.5 → 72.3	45.0 → 57.5
RepViT-M0.9	6.9M	45.8 → 65.2	36.1 → 52.0
Hybrid			
MobileViTv2-100	6.8M	47.0 → 66.2	43.3 → 55.4
MobileViTv2-150	13.1M	43.0 → 71.0	35.7 → 55.5
EdgeNeXt-XXS	2.9M	45.6 → 67.9	37.6 → 53.3
EdgeNeXt-XS	4.0M	48.5 → 68.8	40.7 → 54.7

TABLE VIII

DISTILLATION DESIGN ABLATION ON 6-SHOT SETTING.

ConvNeXt-Atto (CNN)	Cataract101	Cholec80
Student only (w/o distillation)	50.2	39.8
+ L_{cap} (CAP only)	62.4 (+12.2)	47.0 (+7.2)
+ L_{stgd} (STGD only)	59.0 (+8.8)	44.8 (+5.0)
+ L_{cap} + L_{stgd} (Full method)	71.6 (+21.4)	52.2 (+12.4)

notations, showing that our end-to-end spatiotemporal design keeps accurate recognition even under the limited supervision.

2) *Spatiotemporal Module Visualization*: Figure 6 visualizes the top-3 frames that KPTF selects within the input window. The selected frames consistently capture informative moments such as instrument-tissue interaction, while background-dominated views (e.g., out-of-focus or instrument-withdrawn scenes) receive low scores. This shows that KPTF picks discriminative keyframes without explicit supervision. Figure 7 visualizes the spatial token selection of SSTS, where high-scoring patches concentrate on surgical instruments and tissue surfaces rather than on irrelevant background. These demonstrate that **SPR-Teacher** focuses on the spatial and temporal cues that carry the strongest discriminative evidence.

V. CONCLUSION AND LIMITATIONS

Surgical phase recognition is a fundamental task for intelligent operating rooms. In the foundation model era, its clinical deployment is hindered by two deficiencies of the FM paradigm itself: a reliance on massive training data and an inability to infer in real time on clinical hardware. We argue that the dual redundancy of surgical videos—dominant background regions and repetitive steady-state frames—is the key obstacle behind the data deficiency, and propose to combat it in both the model and its distillation. **SPR-Teacher** compresses spatial redundancy through salient spatial token selection and temporal redundancy through keyframe-prior temporal fusion, adapting a frozen generic foundation model with only a handful of labeled videos. **SPR-Lite** then lifts the real-time

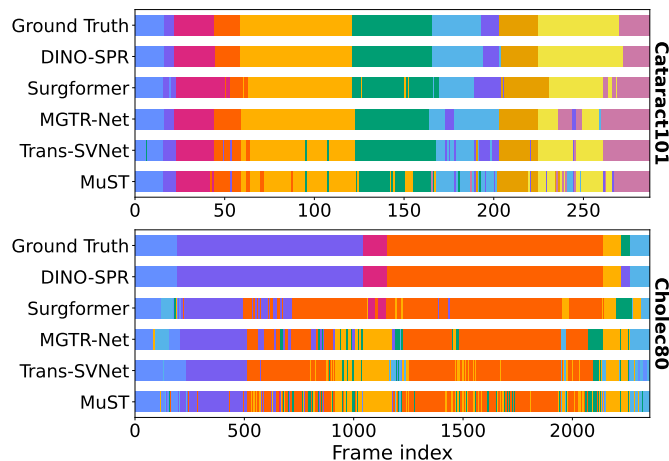


Fig. 5. Per-frame phase predictions of **SPR-Teacher** versus prior methods under the 6-shot setting on Cholec80 and Cataract101.

TABLE IX

CONFIGURATIONS OF THE COMPARED FOUNDATION MODELS.

Method	Backbone	Paradigm	Modality	Data
General FMs				
DINOv3	ViT	DINO	I	1.7B images
CLIP	ViT	CTR	I+T	400M images
VideoMAE2	ViT	MAE	V	1.35M videos
VJEPa2	ViT	JEPa	V	2M videos
InternVideo2	ViT	MAE+CTR	V+T	1.1M videos
Surgical FMs				
EndoSSL	ViT	MSN	I	23.3M images
SurgeNetXL	CAFormer	DINO	I	4.7M images
EndoViT	ViT	MAE	I	744K images
SurgMotion	ViT	JEPa	V	15M videos
EndoFM	ViT	MAE	V	33K videos
SurgVISTA	ViT	MAE	V	3.7K videos
PeskaVLP	ResNet+BERT	CTR	V+T	1.3K videos
HecVL	ResNet+BERT	CTR	V+T	1.3K videos
SurgVLP	ResNet+BERT	CTR	V+T	1.3K videos

I/V/T denotes image/video/text. MAE, DINO, JEPa, and MSN belong to self-supervised learning, while CTR is contrastive learning.

deficiency by compressing the model itself, where a cross-attention projection bridges arbitrary student architectures and a spatiotemporally-guided distillation transfers the teacher’s where- and when-to-look priors as dense supervision. Across four public and one private dataset, **SPR-Teacher** surpasses prior methods under few-shot settings by large margins, while **SPR-Lite** retains this performance with under 9M parameters at over 25 FPS on a single CPU core, offering a practical route toward low-annotation, real-time deployment. Nevertheless, several limitations remain. First, we only consider foundation models with publicly released weights, so the coverage may not be exhaustive. Second, surgical FMs inevitably pre-train on public surgical videos, part of which may overlap with our evaluation data and inflate their scores, making the FM comparison not perfectly fair. Third, real-time efficiency is measured on a server CPU core rather than on clinical hardware, where quantization and further engineering optimization remain to be explored. We leave these for future work.

APPENDIX I

CONFIGURATIONS OF THE COMPARED FOUNDATION MODELS

REFERENCES

- [1] C. Vermeulen, M. Pagès-Gallego, L. Kester, M. Kranendonk, P. Wesseling, N. Verburg, P. de Witt Hamer, E. Kooi, L. Dankmeijer, J. van der Lugt *et al.*, “Ultra-fast deep-learned cns tumour classification during surgery,” *Nature*, vol. 622, no. 7984, pp. 842–849, 2023.
- [2] R. H. Mithany, S. Aslam, S. Abdallah, M. Abdelmaseeh, F. Gerges, M. S. Mohamed, M. Manasseh, A. Wanees, M. H. Shahid, M. S. Khalil *et al.*, “Advancements and challenges in the application of artificial intelligence in surgical arena: a literature review,” *Cureus*, vol. 15, no. 10, 2023.
- [3] J. Cao, H.-C. Yip, Y. Chen, M. Scheppach, X. Luo, H. Yang, M. K. Cheng, Y. Long, Y. Jin, P. W.-Y. Chiu *et al.*, “Intelligent surgical workflow recognition for endoscopic submucosal dissection with real-time animal study,” *Nature Communications*, vol. 14, no. 1, p. 6676, 2023.
- [4] D. Batić, F. Holm, E. Özsoy, T. Czempliel, and N. Navab, “Endovit: pretraining vision transformers on a large collection of endoscopic images,” *International Journal of Computer Assisted Radiology and Surgery*, vol. 19, no. 6, pp. 1085–1091, 2024.

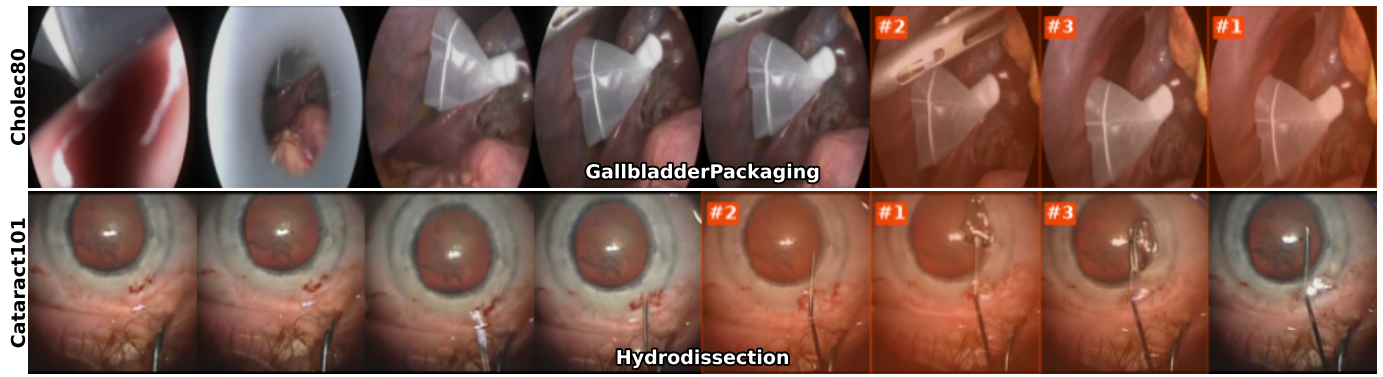


Fig. 6. Keyframe selection of the KPTF module. Within an 8-frame window, the top-3 frames with the largest temporal weights are highlighted (#1–#3); KPTF prioritizes frames with salient instrument-tissue interactions, which carry the strongest discriminative evidence.

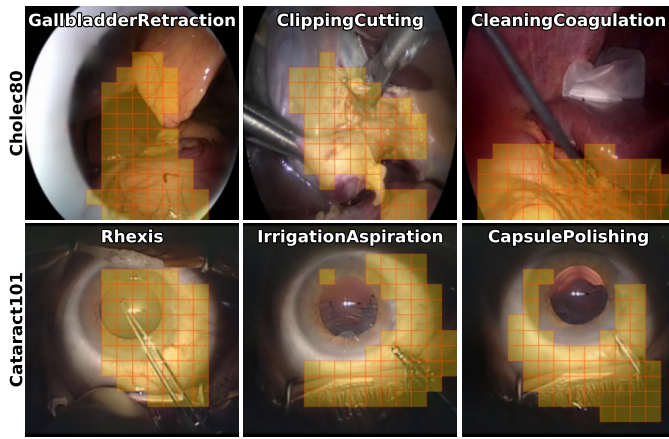


Fig. 7. Spatial token selection of the SSTS module. Selected tokens (yellow) focus on instrument-tissue interactions while suppressing background, revealing the saliency prior guiding phase discrimination.

- [5] S. Yang, F. Zhou, L. Mayer, F. Huang, Y. Chen, Y. Wang, S. He, Y. Nie, X. Wang, Y. Jin *et al.*, “Large-scale self-supervised video foundation model for intelligent surgery,” *npj Digital Medicine*, vol. 9, no. 1, p. 220, 2026.
- [6] J. Wu, F. Holm, C. Chen, A. Wang, Y. Hu, X. Ye, Z. Zang, M. Xu, L. Zhou, H. Liao *et al.*, “Surgmotion: A video-native foundation model for universal understanding of surgical videos,” *arXiv preprint arXiv:2602.05638*, 2026.
- [7] U. Klank, N. Padoy, H. Feussner, and N. Navab, “Automatic feature generation in endoscopic images,” *International Journal of Computer Assisted Radiology and Surgery*, vol. 3, no. 3, pp. 331–339, 2008.
- [8] N. Padoy, T. Blum, S.-A. Ahmadi, H. Feussner, M.-O. Berger, and N. Navab, “Statistical modeling and recognition of surgical workflow,” *Medical image analysis*, vol. 16, no. 3, pp. 632–641, 2012.
- [9] A. P. Twinanda, S. Shehata, D. Mutter, J. Marescaux, M. De Mathelin, and N. Padoy, “Endonet: a deep architecture for recognition tasks on laparoscopic videos,” *IEEE transactions on medical imaging*, vol. 36, no. 1, pp. 86–97, 2016.
- [10] G. Yengera, D. Mutter, J. Marescaux, and N. Padoy, “Less is more: Surgical phase recognition with less annotations through self-supervised pre-training of cnn-lstm networks,” *arXiv preprint arXiv:1805.08569*, 2018.
- [11] Y. Jin, Y. Long, C. Chen, Z. Zhao, Q. Dou, and P.-A. Heng, “Temporal memory relation network for workflow recognition from surgical video,” *IEEE Transactions on Medical Imaging*, vol. 40, no. 7, pp. 1911–1923, 2021.
- [12] Y. Jin, Q. Dou, H. Chen, L. Yu, J. Qin, C.-W. Fu, and P.-A. Heng, “Sv-rnet: workflow recognition from surgical videos using recurrent convolutional network,” *IEEE transactions on medical imaging*, vol. 37, no. 5, pp. 1114–1126, 2017.
- [13] T. Czempel, M. Paschali, M. Keicher, W. Simson, H. Feussner, S. T. Kim, and N. Navab, “Tecno: Surgical phase recognition with multi-stage temporal convolutional networks,” in *International conference on medical image computing and computer-assisted intervention*. Springer, 2020, pp. 343–352.
- [14] D. Kitaguchi, N. Takeshita, H. Matsuzaki, T. Igaki, H. Hasegawa, and M. Ito, “Development and validation of a 3-dimensional convolutional neural network for automatic surgical skill assessment based on spatiotemporal video analysis,” *JAMA network open*, vol. 4, no. 8, pp. e2120786–e2120786, 2021.
- [15] B. Zhang, A. Ghanem, A. Simes, H. Choi, and A. Yoo, “Surgical workflow recognition with 3dcnn for sleeve gastrectomy,” *International Journal of Computer Assisted Radiology and Surgery*, vol. 16, no. 11, pp. 2029–2036, 2021.
- [16] X. Gao, Y. Jin, Y. Long, Q. Dou, and P.-A. Heng, “Trans-svnet: Accurate phase recognition from surgical videos via hybrid embedding aggregation transformer,” in *International conference on medical image computing and computer-assisted intervention*. Springer, 2021, pp. 593–603.
- [17] Y. Liu, J. Huo, J. Peng, R. Sparks, P. Dasgupta, A. Granados, and S. Ourselin, “Skit: a fast key information video transformer for online surgical phase recognition,” in *2023 IEEE/CVF international conference on computer vision (ICCV)*. IEEE, 2023, pp. 21 017–21 027.
- [18] Y. Liu, M. Boels, L. C. Garcia-Peraza-Herrera, T. Vercauteren, P. Dasgupta, A. Granados, and S. Ourselin, “Lovit: Long video transformer for surgical phase recognition,” *Medical Image Analysis*, vol. 99, p. 103366, 2025.
- [19] S. Yang, L. Luo, Q. Wang, and H. Chen, “Surgformer: Surgical transformer with hierarchical temporal attention for surgical phase recognition,” in *International Conference on Medical Image Computing and Computer-Assisted Intervention*. Springer, 2024, pp. 606–616.
- [20] O. Siméoni, H. V. Vo, M. Seitzer, F. Baldassarre, M. Oquab, C. Jose, V. Khalidov, M. Szafraniec, S. Yi, M. Ramamonjisoa *et al.*, “Dinov3,” *arXiv preprint arXiv:2508.10104*, 2025.
- [21] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark *et al.*, “Learning transferable visual models from natural language supervision,” in *International conference on machine learning*. PMLR, 2021, pp. 8748–8763.
- [22] L. Wang, B. Huang, Z. Zhao, Z. Tong, Y. He, Y. Wang, Y. Wang, and Y. Qiao, “Videomae v2: Scaling video masked autoencoders with dual masking,” in *2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 2023, pp. 14 549–14 560.
- [23] M. Assran, A. Bardes, D. Fan, Q. Garrido, R. Howes, M. Muckley, A. Rizvi, C. Roberts, K. Sinha, A. Zholus *et al.*, “V-jepa 2: Self-supervised video models enable understanding, prediction and planning,” *arXiv preprint arXiv:2506.09985*, 2025.
- [24] Y. Wang, K. Li, X. Li, J. Yu, Y. He, G. Chen, B. Pei, R. Zheng, Z. Wang, Y. Shi *et al.*, “Internvideo2: Scaling foundation models for multimodal video understanding,” in *European conference on computer vision*. Springer, 2024, pp. 396–416.
- [25] P. Dermeyer, A. Kalra, and M. Schwartz, “Endodino: A foundation model for gi endoscopy,” *arXiv preprint arXiv:2501.05488*, 2025.
- [26] R. de Jong, H. Carolus, H. Franciscus, R. C. van Jaarsveld, R. van Hilleberg, P. Josien, P. H. de With, Y. al Khalil, F. van Der Sommen *et al.*, “Scaling up self-supervised learning for improved surgical foundation models,” *Medical Image Analysis*, p. 103873, 2025.
- [27] C. Che, C. Wang, T. Vercauteren, S. Tsoka, and L. C. Garcia-Peraza-

- Herrera, "Surg-3m: A dataset and foundation model for perception in surgical settings," *arXiv e-prints*, pp. arXiv-2503, 2025.
- [28] R. Hirsch, M. Caron, R. Cohen, A. Livne, R. Shapiro, T. Golany, R. Goldenberg, D. Freedman, and E. Rivlin, "Self-supervised learning for endoscopic video analysis," in *International conference on medical image Computing and computer-assisted intervention*. Springer, 2023, pp. 569–578.
- [29] Z. Wang, C. Liu, S. Zhang, and Q. Dou, "Foundation model for endoscopy video analysis via large-scale self-supervised pre-train," in *International Conference on Medical Image Computing and Computer-Assisted Intervention*. Springer, 2023, pp. 101–111.
- [30] Z. Wang, C. Liu, L. Zhu, T. Wang, S. Zhang, and Q. Dou, "Improving foundation model for endoscopy video analysis via representation learning on long sequences," *IEEE Journal of Biomedical and Health Informatics*, 2025.
- [31] K. Yuan, V. Srivastav, T. Yu, J. L. Lavanchy, J. Marescaux, P. Mascagni, N. Navab, and N. Padoy, "Learning multi-modal representations by watching hundreds of surgical video lectures," *Medical Image Analysis*, p. 103644, 2025.
- [32] K. Yuan, V. Srivastav, N. Navab, and N. Padoy, "Hecvl: Hierarchical video-language pretraining for zero-shot surgical phase recognition," in *International Conference on Medical Image Computing and Computer-Assisted Intervention*. Springer, 2024, pp. 306–316.
- [33] K. Yuan, N. Navab, N. Padoy *et al.*, "Procedure-aware surgical video-language pretraining with hierarchical knowledge augmentation," *Advances in Neural Information Processing Systems*, vol. 37, pp. 122 952–122 983, 2024.
- [34] M. Hu, K. Yuan, Y. Shen, F. Tang, X. Xu, L. Zhou, W. Li, Y. Chen, Z. Xu, Z. Peng *et al.*, "Ophclip: Hierarchical retrieval-augmented learning for ophthalmic surgical video-language pretraining," *arXiv preprint arXiv:2411.15421*, 2024.
- [35] Y. Wanyan, X. Yang, W. Dong, and C. Xu, "A comprehensive review of few-shot action recognition," *arXiv preprint arXiv:2407.14744*, 2024.
- [36] D. Ferber, G. Wölflein, I. C. Wiest, M. Liger, S. Sainath, N. G. Laleh, O. S. M. El Nahhas, G. Müller-Franzes, D. Jäger, D. Truhn, and J. N. Kather, "In-context learning enables multimodal large language models to classify cancer pathology images," *Nat. Commun.*, vol. 15, p. 10104, 2024.
- [37] R. Dutt, L. Ericsson, P. Sanchez, S. A. Tsafaris, and T. Hospedales, "Parameter-efficient fine-tuning for medical image analysis: The missed opportunity," in *Proc. Med. Image Understanding Anal. (MIDL)*, 2024.
- [38] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen, "Lora: Low-rank adaptation of large language models," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2022.
- [39] B. Bergner, C. Lippert, and A. Mahendran, "Token Cropr: Faster ViTs for quite a few tasks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2025.
- [40] Z. Liu, K. Xu, B. Su, X. Zou, Y. Peng, and J. Zhou, "STOP: Integrated spatial-temporal dynamic prompting for video understanding," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2025.
- [41] G. Hinton, O. Vinyals, and J. Dean, "Distilling the knowledge in a neural network," *arXiv preprint arXiv:1503.02531*, 2015.
- [42] R. Vemulapalli, H. Pouransari, F. Faghri, S. Mehta, M. Farajtabar, M. Rastegari, and O. Tuzel, "Knowledge transfer from vision foundation models for efficient training of small task-specific models," in *ICML*, 2024.
- [43] J. Lee, D. Das, M. Hayat, S. Choi, K. Hwang, and F. Porikli, "Customkd: Customizing large vision foundation for edge model improvement via knowledge distillation," in *2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 2025, pp. 25 176–25 186.
- [44] J. Fan, C. Li, X. Liu, and A. Yao, "Scalekd: Strong vision transformers could be excellent teachers," *Advances in Neural Information Processing Systems*, vol. 37, pp. 63 290–63 315, 2024.
- [45] Y. Liu, J. Cao, B. Li, W. Hu, J. Ding, and L. Li, "Cross-architecture knowledge distillation," in *Proceedings of the Asian conference on computer vision*, 2022, pp. 3396–3411.
- [46] R. Pei, J. Liu, W. Li, B. Shao, S. Xu, P. Dai, J. Lu, and Y. Yan, "Clipping: Distilling clip-based models with a student base for video-language retrieval," in *2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 2023, pp. 18 983–18 992.
- [47] M. Hatano, R. Hachiuma, R. Fujii, and H. Saito, "Multimodal cross-domain few-shot learning for egocentric action recognition," in *European conference on computer vision*. Springer, 2024, pp. 182–199.
- [48] K. Huang, X.-L. Yuan, R.-D. Liu, L.-S. Ye, Y. Zhou, B. Hu, and Z. Yi, "Multi-teacher temporal regulation network for surgical workflow recognition," *IEEE Transactions on Medical Imaging*, vol. 44, no. 11, pp. 4690–4703, 2025.
- [49] A. Pérez, S. Rodríguez, N. Ayobi, N. Aparicio, E. Dessevres, and P. Arbeláez, "Must: Multi-scale t ransformers for surgical phase recognition," in *International conference on medical image computing and computer-assisted intervention*. Springer, 2024, pp. 422–432.